Table 2: **Generation success and computational cost on MaLiang-IBench.** "Token Limit" denotes the subset of failures in which a run exhausts its allocated token budget, potentially due to reasoning stagnation and unbounded agentic loops. "Orange/blue" indicates "higher/lower" values relative to DeepSeek-V4.1-Flash, which do not imply "better/worse" performance given differing success rates.

| Model | Task Completion | | | Generation Time | | | Model Usage | | |
|---|---|---|---|---|---|---|---|---|---|
| | Success ↑ (%) | Failures ↓ (cases) | Token Limit ↓ (cases) | Total Time (task-h) | Time/Qualified ↓ (min/image) | Time/Task (min/case) | Calls (calls/case) | Input Tokens (k/case) | Output Tokens (k/case) |
| DeepSeek-V4.1-Flash | 12.0 | 44 | 5 | 2.71 | 27.15 | 1.91 | 2.48 | 100.8 | 24.2 |
| DeepSeek-V4-Pro | 18.0 | 41 | 3 | 6.42 | 48.12 | 7.36 | 4.50 | 198.9 | 39.6 |
| Kimi-K2.6 | 34.0 | 33 | 1 | 6.27 | 62.69 | 5.14 | 9.14 | 660.0 | 48.2 |
| Kimi-K2.7-Code | 40.0 | 30 | 1 | 5.97 | 29.84 | 3.59 | 12.94 | 1056.5 | 59.0 |
| Kimi-K3 | 18.0 | 41 | 0 | 5.23 | 34.88 | 5.89 | 2.82 | 308.9 | 17.8 |
| GPT-5.6-Luna | 92.0 | 4 | 0 | 1.65 | 4.49 | 1.98 | 11.10 | 271.6 | 9.3 |
| GPT-5.6-Terra | 92.0 | 4 | 0 | 1.73 | 4.32 | 2.07 | 8.32 | 186.0 | 8.0 |
| GPT-5.6-Sol | 96.0 | 2 | 1 | 2.35 | 3.28 | 2.82 | 6.98 | 209.2 | 13.1 |
| GPT-6-Luna | 96.0 | 2 | 0 | 3.61 | 4.92 | 3.27 | 14.70 | 459.2 | 16.4 |
| GPT-6-Sol | 96.0 | 2 | 0 | 2.70 | 3.52 | 3.24 | 7.42 | 180.1 | 10.2 |
| GPT-6-Astra | 100.0 | 0 | 0 | 2.96 | 3.70 | 3.55 | 6.34 | 142.0 | 8.1 |

Table 3: **Visual quality on MaLiang-IBench.** Quality counts include samples scoring at least 4 on a five-point scale. Means cover successful samples only. Gray cells mark DeepSeek and Kimi results, whose low success rates limit the comparability of mean quality scores. Bold marks the best GPT results. "Blue/orange" denotes "higher/lower" values relative to GPT-5.6-Luna.

| Model | Generation Outcomes | | Quality Threshold Counts | | | Mean Quality Scores | | |
|---|---|---|---|---|---|---|---|---|
| | Successful ↑ (cases) | All Criteria ↑ (cases) | Alignment ↑ (cases) | Aesthetics ↑ (cases) | Composition ↑ (cases) | Alignment ↑ (1–5) | Aesthetics ↑ (1–5) | Composition ↑ (1–5) |
| DeepSeek-V4.1-Flash | 6 | 6 | 6 | 6 | 6 | 4.33 | 4.00 | 4.17 |
| DeepSeek-V4-Pro | 9 | 8 | 8 | 8 | 9 | 4.11 | 3.89 | 4.11 |
| Kimi-K2.6 | 17 | 6 | 7 | 8 | 9 | 3.47 | 3.41 | 3.65 |
| Kimi-K2.7-Code | 20 | 12 | 13 | 13 | 13 | 3.75 | 3.60 | 3.70 |
| Kimi-K3 | 9 | 9 | 9 | 9 | 9 | 4.56 | 4.22 | 4.33 |
| GPT-5.6-Luna | 46 | 22 | 25 | 33 | 40 | 3.57 | 3.74 | 3.93 |
| GPT-5.6-Terra | 46 | 24 | 27 | 31 | 40 | 3.70 | 3.70 | 3.93 |
| GPT-5.6-Sol | 48 | 43 | 43 | 45 | 48 | 4.23 | 4.10 | 4.27 |
| GPT-6-Luna | 48 | 44 | 44 | 48 | 48 | 4.10 | 4.08 | 4.15 |
| GPT-6-Sol | 48 | 46 | 46 | 48 | 48 | 4.25 | 4.15 | 4.25 |
| GPT-6-Astra | **50** | **48** | **48** | **50** | **50** | **4.36** | **4.22** | **4.54** |

generated images from the three GPT-6 models meet the aesthetics and composition thresholds. Their remaining quality failures concern prompt adherence. This pattern shows that successful generation alone is insufficient to assess executable visual programs.

Besides, mean quality scores must be interpreted alongside the number of successful tasks. DeepSeek and Kimi generate only 6-20 images, so their means describe a limited subset of MaLiang-IBench. Kimi-K3's alignment score of 4.56 is based on just nine images and does not establish superior performance over the full evaluation set. Astra obtains the highest mean score in each quality dimension among the GPT models. The cost results reveal a trade-off between generation time and the number of images satisfying all criteria. GPT-5.6-Sol produces 43 qualifying images at 3.28 minutes per image. Astra produces 48 at 3.70 minutes per image. These time estimates include failures and retries and thus account for the cost of obtaining images that meet the quality thresholds.

**Qualitative results on MaLiang-IBench.** Fig. 6 illustrates differences in visual realization across the evaluated models. The greenhouse (last row) and robot ensemble (fourth row) examples reveal differences in subject scale, scene detail, and the depiction of individual objects across models. The teacup (first row) and ginkgo-leaf examples (third row) further probe the integration of flat cartoon characters with realistic materials. Astra's results show more pronounced surface detail and lighting cues in these examples, while the architectural board combines a main view with supporting elevations and construction details. These selected examples complement the aggregate scores by showing how the same prompt can lead to different compositions and levels of visual detail.

**Quantitative results on MaLiang-VBench.** Tables 4 and 5 show a similar gap between generation success and quality on MaLiang-VBench. **Astra satisfies all four quality thresholds on 10 of its 13 successful generations.** GPT-5.6-Sol satisfies all thresholds on five of its seven successful generations. Kimi-K2.6 completes three tasks but none of its videos meets all criteria. DeepSeek-